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Deep thermalization

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- Some examples

MAIN REFERENCES

- Joonhee Choi, Adam L. Shaw, Ivaylo S. Madjarov, Xin Xie, Ran Finkelstein, Jacob P. Covey, Jordan S. Cotler, Daniel K. Mark, Hsin-Yuan Huang, Anant Kale, Hannes Pichler, Fernando G.S.L. Brandão, Soonwon Choi, and Manuel Endres, *Preparing random states and benchmarking with many-body quantum chaos*, Nature **613**, 468 (2023).

- Jordan S. Cotler, Daniel K. Mark, Hsin-Yuan Huang, Felipe Hernández, Joonhee Choi, Adam L. Shaw, Manuel Endres, and Soonwon Choi, *Emergent Quantum State Designs from Individual Many-Body Wave Functions*, PRX Quantum **4**, 010311 (2023).

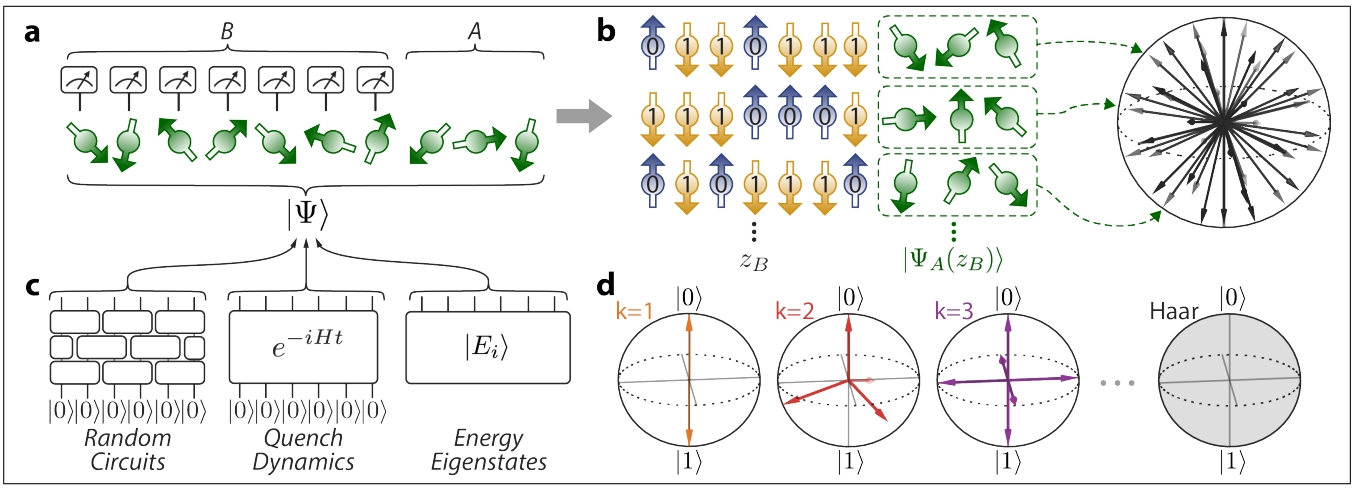
INTRODUCTION (FIGURE)

- **a:** We consider a quantum system in a state $|\Psi\rangle$ decomposed in subsystems A and B. The average outcomes of measurements on A are encoded in the reduced density matrix $\mathrm{Tr}_B(\rho)$. The main idea is to go beyond averages and to consider *distributions* of measurement outcomes. For this, it is supposed that one can measure the state of the *full* system.

- **b:** For each measurement outcome z_B of subsystem B, a state $|\Psi_A(z_B)\rangle$ results. The number of such states is given $\dim(\mathcal{H}_B)$. In this way, one constructs an *projected ensemble* $\mathcal{E} = \{p(z_B)|\Psi_A(z_B)\}$. The Bloch sphere picture on the right is for illustrational purposes only, in reality the states $|\psi_A(z_B)\rangle$ are not two-dimensional.

- **c:** Examples of systems that can be used to generate states from which a projected ensemble can be constructed. Importantly, these system does not have to involve randomness.

- **d:** Examples of *quantum state k-designs* of order 1, 2, 3, and ∞. For these ensembles, measurement outcomes agree up to the k-th order with the (maximally random) Haar ensemble. A more detailed introduction follows.



PROJECTED ENSEMBLES

After the measurement, with probability $p(z_B)$, the system is described by the normalized wave function $|\Psi_A(z_B)\rangle \otimes |z_B\rangle$, where

$$p(z_B) = \langle\Psi|\big(\mathbb{1}_A \otimes |z_B\rangle\langle z_B|\big)|\Psi\rangle, \tag{1}$$

$$|\Psi_A(z_B)\rangle = \frac{1}{\sqrt{p(z_B)}}\big(\mathbb{1}_A \otimes \langle z_B|\big)|\Psi\rangle. \tag{2}$$

This defines the *projected ensemble* $\mathcal{E} = \{p(z_B), |\Psi_A(z_B)\rangle\}$. The ensemble consists of $\dim(\mathcal{H}_B)$ states. These states are generally not pairwise orthogonal. For example, non-orthogonality is guaranteed to be the case when $\dim(\mathcal{H}_B) > \dim(\mathcal{H}_A)$.

QUANTUM STATE K-DESIGNS

The k-th moment of an observable $\mathcal{O}$ for an ensemble $\mathcal{E} = \{p_i, \Psi_i\}$ is given by

$$\langle\mathcal{O}^k\rangle = \sum_i p_i\, \mathcal{O}_i^k, \qquad \mathcal{O}_i = \langle\Psi_i|\mathcal{O}|\Psi_i\rangle. \tag{3}$$

In a more convenient notation:

$$\langle\mathcal{O}^k\rangle = \mathrm{Tr}\left(\mathcal{O}^{\otimes k}\rho_{\mathcal{E}}^{(k)}\right), \qquad \rho_{\mathcal{E}}^{(k)} = \sum_i p_i\,(|\Psi_i\rangle\langle\Psi_i|)^{\otimes k}. \tag{4}$$

The randomness of an ensemble of states is quantified through its k-th moments. An ensemble is called a *quantum state k-design* if its k-th moment equals the k-th moment of the Haar-random ensemble:

$$\rho_{\mathcal{E}}^{(k)} = \rho_{\mathrm{Haar}}^{(k)}. \tag{5}$$

The construction of $\rho_{\mathrm{Haar}}^{(k)}$ is discussed in more detail below. For a quantum state k-design, the distribution of measurement outcomes matches these of the Haar ensemble up to (and including) the k-th moment.

MOMENTS OF THE HAAR ENSEMBLE

The Haar ensemble is induced by the *Haar measure* $d\mu_{\text{Haar}}$. For any unitary matrix V, the Haar measure over states $|\Psi\rangle$ satisfies

$$d\mu_{\text{Haar}}(|\Psi\rangle) = d\mu_{\text{Haar}}(V|\Psi\rangle), \tag{6}$$

which fixes the measure uniquely. Then,

$$\rho^{(k)}_{\text{Haar}} = \int d\mu_{\text{Haar}}(|\Psi\rangle)\big(\Psi\rangle\langle\Psi|\big)^{\otimes k}. \tag{7}$$

For $k = 1$, one gets $\rho^{(1)}_{\text{Haar}} = N^{-1}\mathbb{1}$. Physically, this says that the expectation value $\langle\mathcal{O}\rangle_{\text{Haar}}$ of an operator $\mathcal{O}$ averaged over the Haar ensemble obeys

$$\langle\mathcal{O}\rangle_{\text{Haar}} = \text{Tr}\left[\rho^{(1)}_{\text{Haar}}\mathcal{O}\right] \tag{8}$$

$$= \frac{1}{N}\,\text{Tr}[\mathcal{O}], \tag{9}$$

i.e. one finds the infinite-temperature average (as intuitively expected).

In their numerics, the authors approximate $\rho^{(k)}_{\text{Haar}}$ by the empirical distribution $\rho^{(k)}_{\text{em}}$ obtained as

$$\rho^{(k)}_{\text{em}} = \frac{1}{2^{N_B}}\sum_{i=1}^{2^{N_B}}(|\Psi_i\rangle\langle\Psi_i|)^{\otimes k}, \qquad |\Psi_i\rangle \sim \text{Haar}(d). \tag{10}$$

The $|\Psi_r\rangle$ can be obtained as eigenstates of random matrices for which the eigenvectors are Haar distributed. An example is the Gaussian unitary ensemble. An $N \times N$ matrix M sampled from this ensemble can be constructed as

$$M = \frac{1}{2}(A + A^\dagger), \tag{11}$$

where A is an $N \times N$ matrix with complex-valued elements $A_{nm} = u_{nm} + iv_{nm}$ with u_{nm} and v_{nm} sampled independently from the normal distribution with mean zero and variance 1/2.

It is known how $\rho^{(k)}_{\text{Haar}}$ can be construced exactly (the appendix A of the PRX paper cited above). However, the procedure is computationally expensive. For small dimensions and k, it is observed that

$$\mathbb{E}_{\Psi_1,\Psi_2,\ldots\sim\text{Haar}(d)}\left[\frac{1}{2}||\rho^{(k)}_{\text{em}} - \rho^{(k)}_{\text{Haar}}||_1\right] \sim \frac{1}{\sqrt{2^{N_B}}}. \tag{12}$$

The trace distance $||A||_1$ is defined as

$$||A||_1 = \text{Tr}[\sqrt{A^\dagger A}], \tag{13}$$

which is (in the current context) related to the probability of distinguishing $\rho^{(k)}_{\text{em}}$ and $\rho^{(k)}_{\text{Haar}}$ under measurements.

I. SOME EXAMPLES

Example: single qubit with equal probabilities for $\{|0\rangle, |1\rangle\}$, which is a quantum state 1-design:

$$\rho^{(1)} = \frac{1}{2}\Big(|0\rangle\langle 0| + |1\rangle\langle 1|\Big) \tag{14}$$

$$= \rho^{(1)}_{\text{Haar}}. \tag{15}$$

It is not a quantum state 2-design as

$$\rho^{(2)} = \frac{1}{2}\Big(|0\rangle\langle 0| \otimes |0\rangle\langle 0| + |1\rangle\langle 1| \otimes |1\rangle\langle 1|\Big) \tag{16}$$

$$\neq \rho^{(2)}_{\text{Haar}}. \tag{17}$$

To form a 2-design for a single qubit, one can use four distinct nonorthogonal quantum states uniformly spread over the Bloch sphere.

Example: one-dimensional quantum Ising spin system with mixed fields. The Hamiltonian H is given by

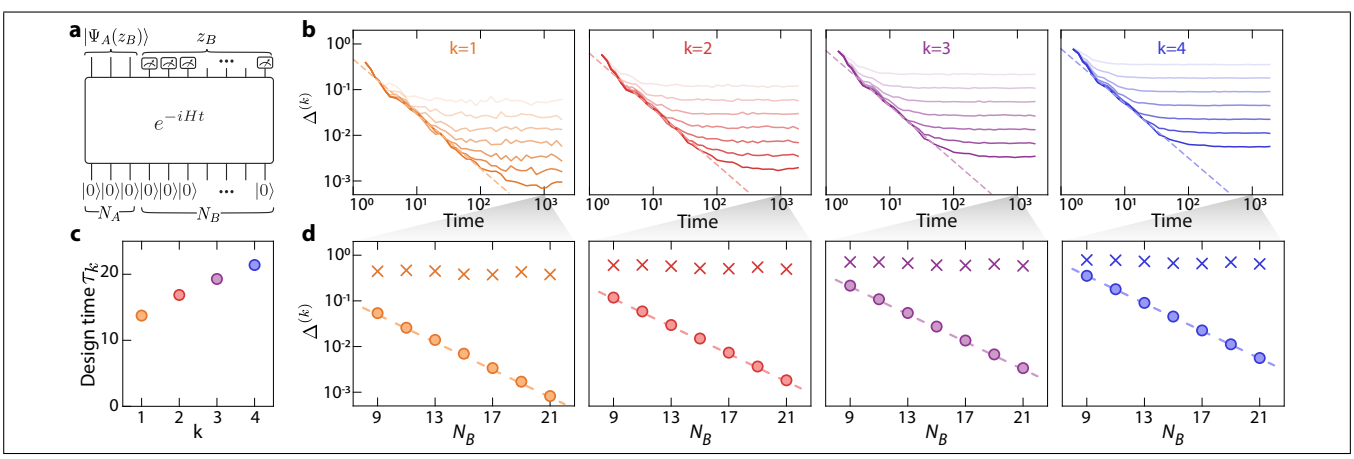
$$H = h^x \sum_{j=1}^{N} \sigma_j^x + h^y \sum_{j=1}^{N} \sigma_j^y + J \sum_{j=1}^{N-1} \sigma_j^x \sigma_{j+1}^x. \tag{18}$$

The parameters are taken as $h^x = 0.8090$, $h^y = 0.9045$, and $J = 1$ (chosen to match some experimental parameters). For these parameters, the eigenstate thermalization hypothesis has been verified (Kim, Ikeda, Huse, PRE 2014). The system is initialized in an infinite-temperature (zero energy expectation value) product state.

The quantity of interest $\Delta^{(k)}$ is given by the trace distance between $\rho^{(k)}$ and $\rho^{(k)}_{\text{Haar}}$,

$$\Delta^{(k)} = \frac{1}{2}||\rho^{(k)} - \rho^{(k)}_{\text{Haar}}||_1, \tag{19}$$

which is again related to the probability of distinguishing them under measurements.



Empirical observation: $\Delta^{(k)}(t) \sim t^{-1.2}$. Since the empirical ensemble asymptotically approaches the Haar ensemble in the limit of infinite samples, $\Delta^{(k)}$ is determined only by statistical fluctuations associated with having a finite number of quantum states. The Haar ensemble is samled 2^{N_B} times.

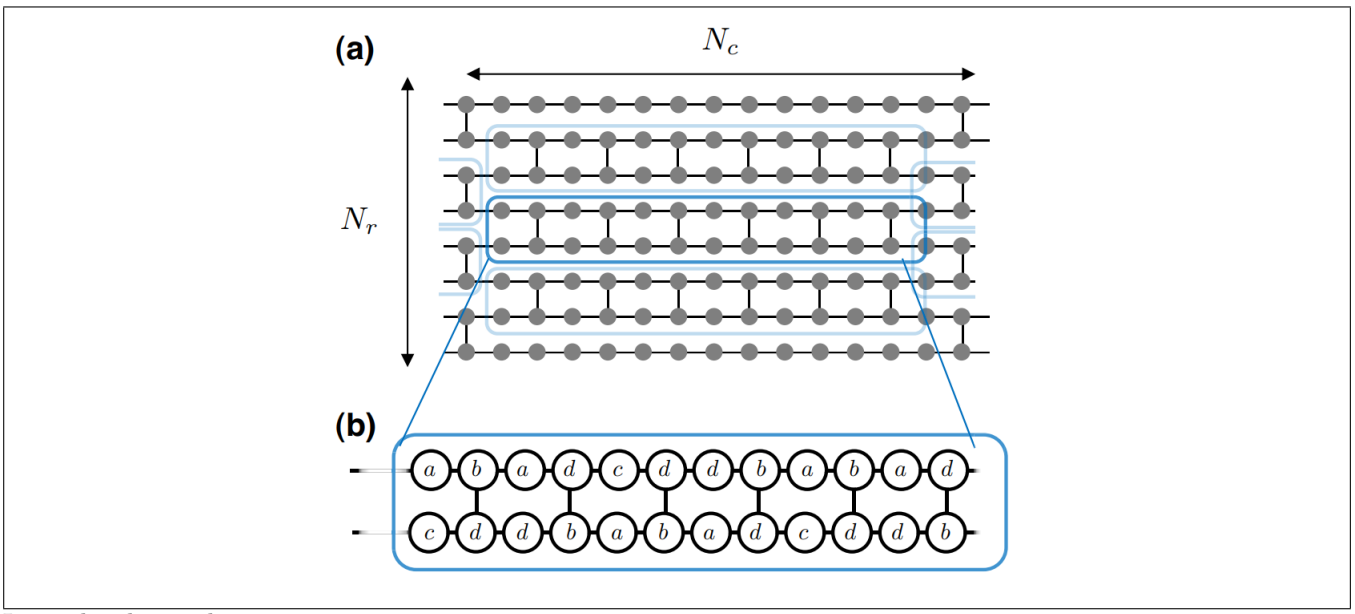
The two main questions in the field can be summarized as follows:

- Which systems obey deep thermalization?
- How long do they to do so?

Reverse perspective: the above finding establishes that one can (experimentally) generate Haar-random states from single time-evolved states. A system for which this can be shown analytically is the Hamiltonian

$$H_{\text{Ising}} = \sum_i h_i \sigma_i^x + \sum_{(i,j)\in E} J_0 \sigma_i^x \sigma_j^x \tag{20}$$

for duration $\tau = \pi/(4J_0)$ prepared in the initial state $|\Psi_0\rangle = |0\rangle^{\otimes N}$. Here, the on-site field strength takes one of the four values $h_a = 0$, $h_b = J_0/2$, $h_c = -J_0/2$, $h_d = J_0$, depending on which group $E = \{a, b, c, d\}$the qubit at site i belongs to (figure below). The Ising interactions with strength J_0 are present only for nearest-neighboring qubit pairs connected by edges of the graph.



It can be shown that

$$\left\| \rho_{\mathcal{E}}^{(k)} - \rho_{\text{Haar}}^{(k)} \right\|_1 \leq \epsilon \tag{21}$$

if

$$N_c \geq C(N_r k + \log_2(1/\epsilon)) \tag{22}$$

for some constant C (Mezher, Ghalbouni, Dgheim, Markham, PRA 018). Note that here the size of subsystems A and B does not enter. The point to realize is that physical systems typically obey deep thermalization only up to a finite precision.